

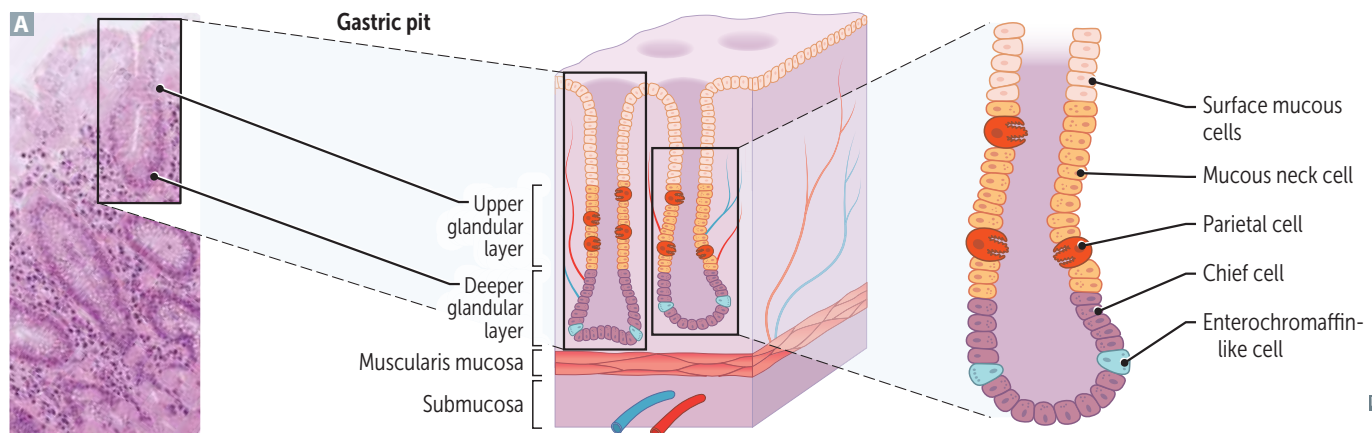
► GASTROINTESTINAL—PHYSIOLOGY

Gastrointestinal regulatory substances

REGULATORY SUBSTANCE	SOURCE	ACTION	REGULATION	NOTES
Gastrin	G cells (antrum of stomach, duodenum)	↑ gastric H ⁺ secretion ↑ growth of gastric mucosa ↑ gastric motility	↑ by stomach distention/ alkalinization, amino acids, peptides, vagal stimulation via gastrin-releasing peptide (GRP) ↓ by pH < 1.5	↑ by chronic PPI use ↑ in chronic atrophic gastritis (eg, <i>H pylori</i>) ↑↑ in Zollinger-Ellison syndrome (gastrinoma)
Ghrelin	Stomach	↑ appetite (“ghrowlin’ stomach”)	↑ in fasting state ↓ by food	↑ in Prader-Willi syndrome ↓ after gastric bypass surgery
Somatostatin	D cells (pancreatic islets, GI mucosa)	↓ gastric acid and pepsinogen secretion ↓ pancreatic and small intestine fluid secretion ↓ gallbladder contraction ↓ insulin and glucagon release	↑ by acid ↓ by vagal stimulation	Inhibits secretion of various hormones (encourages somato-stasis) Octreotide is an analog used to treat acromegaly, carcinoid syndrome, VIPoma, and variceal bleeding
Cholecystokinin	I cells (duodenum, jejunum)	↑ pancreatic secretion ↑ gallbladder contraction ↓ gastric emptying ↑ sphincter of Oddi relaxation	↑ by fatty acids, amino acids	Acts on neural muscarinic pathways to cause pancreatic secretion
Secretin	S cells (duodenum)	↑ pancreatic HCO ₃ [−] secretion ↓ gastric acid secretion ↑ bile secretion	↑ by acid, fatty acids in lumen of duodenum	↑ HCO ₃ [−] neutralizes gastric acid in duodenum, allowing pancreatic enzymes to function
Glucose-dependent insulintropic peptide	K cells (duodenum, jejunum)	Exocrine: ↓ gastric H ⁺ secretion Endocrine: ↑ insulin release	↑ by fatty acids, amino acids, oral glucose	Also called gastric inhibitory peptide (GIP) Oral glucose load ↑ insulin compared to IV equivalent due to GIP secretion
Motilin	Small intestine	Produces migrating motor complexes (MMCs)	↑ in fasting state	Motilin receptor agonists (eg, erythromycin) are used to stimulate intestinal peristalsis.
Vasoactive intestinal polypeptide	Parasympathetic ganglia in sphincters, gallbladder, small intestine	↑ intestinal water and electrolyte secretion ↑ relaxation of intestinal smooth muscle and sphincters ↓ gastric acid secretion	↑ by distention and vagal stimulation ↓ by adrenergic input	VIPoma —non-α, non-β islet cell pancreatic tumor that secretes VIP; associated with Watery Diarrhea, Hypokalemia, Achlorhydria (WDHA syndrome)
Nitric oxide		↑ smooth muscle relaxation, including lower esophageal sphincter (LES)		Loss of NO secretion is implicated in ↑ LES tone of achalasia

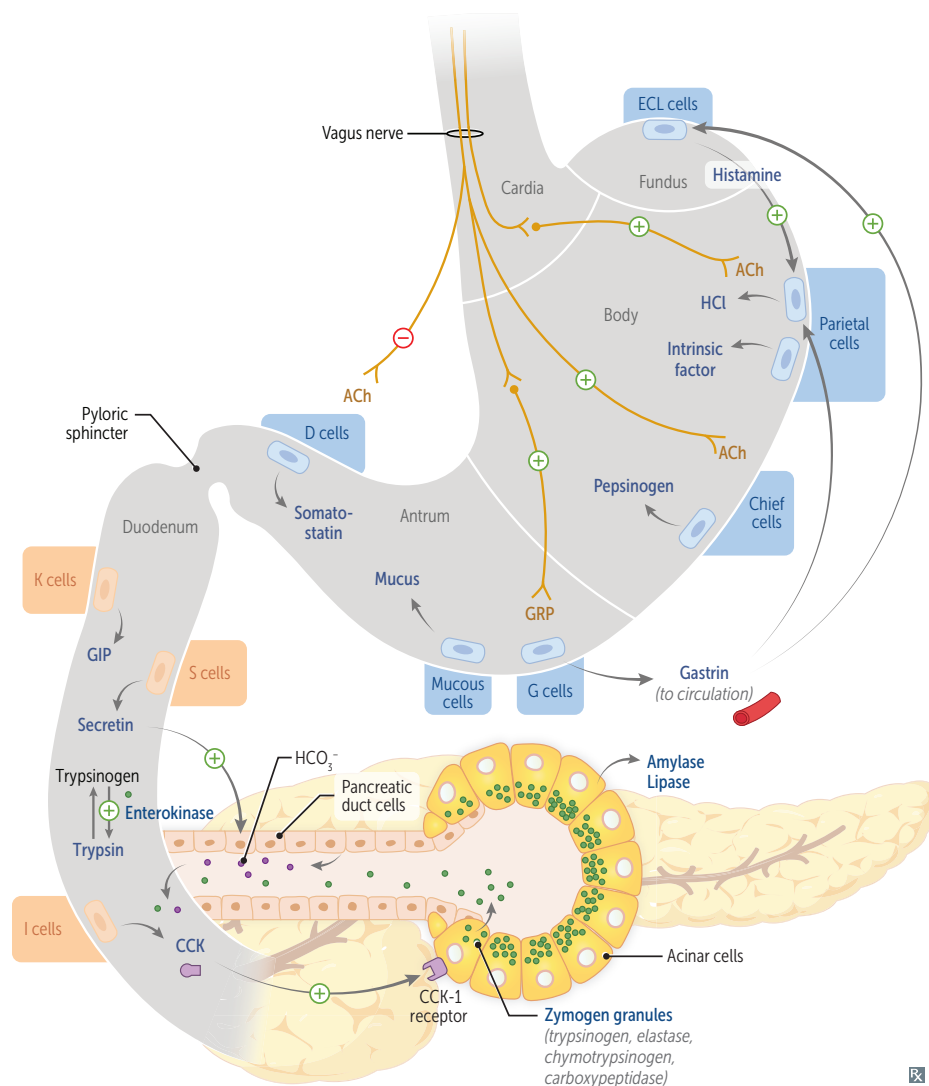
Gastrointestinal secretory products

PRODUCT	SOURCE	ACTION	REGULATION	NOTES
Gastric acid	Parietal cells (stomach A)	↓ stomach pH	↑ by histamine, vagal stimulation (ACh), gastrin ↓ by somatostatin, GIP, prostaglandin, secretin	Autoimmune destruction of parietal cells (pink/eosinophilic histology) → chronic gastritis and pernicious anemia
Intrinsic factor	Parietal cells (stomach)	Vitamin B ₁₂ -binding protein (required for B ₁₂ uptake in terminal ileum)		
Pepsin	Chief cells (stomach)	Protein digestion	↑ by vagal stimulation (ACh), local acid	Pepsinogen (inactive) is converted to pepsin (active) in the presence of H ⁺
Bicarbonate	Mucosal cells (stomach, duodenum, salivary glands, pancreas) and Brunner glands (duodenum)	Neutralizes acid	↑ by pancreatic and biliary secretion with secretin	Trapped in mucus that covers the gastric epithelium



Locations of gastrointestinal secretory cells

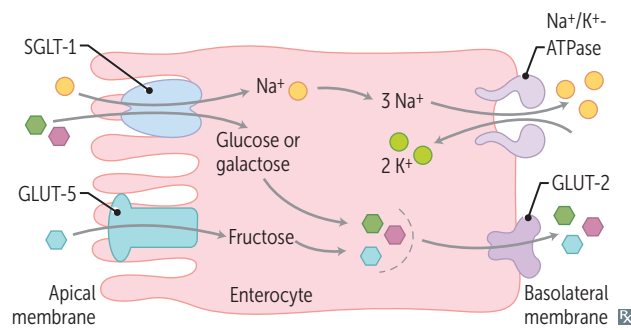
Gastrin ↑ acid secretion primarily through its effects on enterochromaffin-like (ECL) cells (leading to histamine release) rather than through its direct effect on parietal cells.



Pancreatic secretions

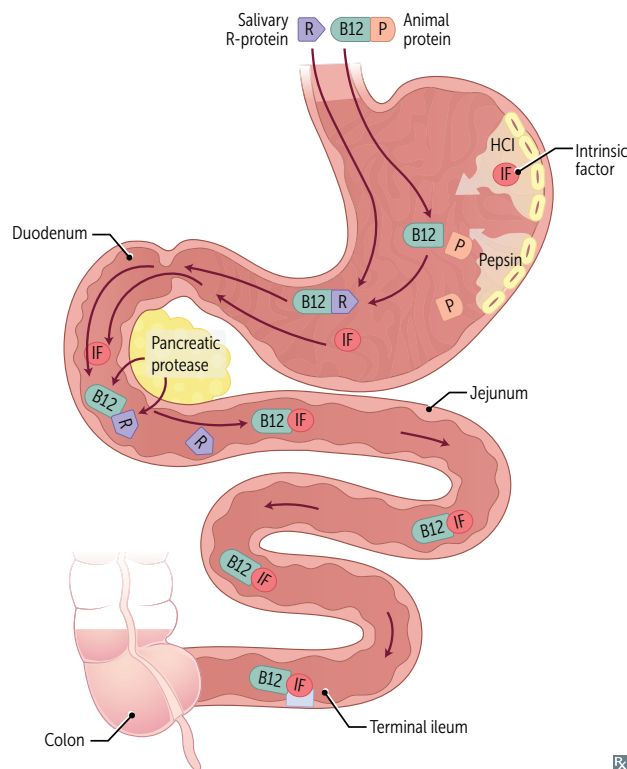
Isotonic fluid; low flow → high Cl^- , high flow → high HCO_3^- .

ENZYME	ROLE	NOTES
α-amylase	Starch digestion	Secreted in active form
Lipases	Lipid digestion	
Proteases	Protein digestion	Includes trypsin, chymotrypsin, elastase, carboxypeptidases Secreted as proenzymes also called zymogens Dipeptides and tripeptides degraded within intestinal mucosa via intracellular process
Trypsinogen	Converted to active enzyme trypsin → activation of other proenzymes and cleaving of additional trypsinogen molecules into active trypsin (positive feedback loop)	Converted to trypsin by enterokinase/enteropeptidase, a brush-border enzyme on duodenal and jejunal mucosa

Carbohydrate absorption

Only monosaccharides (glucose, galactose, fructose) are absorbed by enterocytes. Glucose and galactose are taken up by SGLT1 (Na^+ dependent). Fructose is taken up via facilitated diffusion by GLUT5. All are transported to blood by GLUT2.

D-xylose test: simple sugar that is passively absorbed in proximal small intestine; blood and urine levels ↓ with mucosal damage, normal in pancreatic insufficiency.

Vitamin and mineral absorption

Vitamin and mineral deficiencies may develop in patients with small bowel disease, bowel resection, intestinal failure (also called short bowel syndrome), or bariatric surgery (eg, vitamin B_{12} deficiency complicating terminal ileum resection requires supplementation).

Iron absorbed as Fe^{2+} in duodenum.

Folates absorbed in duodenum and jejunum.

Vitamin B_{12} absorbed in terminal ileum along with bile salts, requires intrinsic factor.

Iron fist, Bro

Dumping syndrome—hyperosmolar food (often sugary) moves too quickly from the stomach to small intestine. Typically occurs after stomach or esophageal surgery.

Peyer patches

Unencapsulated lymphoid tissue **A** found in lamina propria and submucosa of ileum. Contain specialized **M**icrofold (**M**) cells that sample and present antigens to **i**mmune cells. B cells stimulated in germinal centers of Peyer patches differentiate into IgA-secreting plasma cells, which ultimately reside in lamina propria. IgA receives protective secretory component and is then transported across the epithelium to the gut to deal with intraluminal antigen.

Think of **IgA**, the **I**ntra-gut **A**ntibody

Bile

Composed of bile salts (bile acids conjugated to glycine or taurine, making them water soluble), phospholipids, cholesterol, bilirubin, water, and ions. Cholesterol 7 α -hydroxylase catalyzes rate-limiting step of bile acid synthesis.

Functions:

- Digestion and absorption of lipids and fat-soluble vitamins
- Bilirubin and cholesterol excretion (body's 1 $^{\circ}$ means of elimination)
- Antimicrobial activity (via membrane disruption)

↓ absorption of enteric bile salts at distal ileum (as in short bowel syndrome, Crohn disease) prevents normal fat absorption and may cause bile acid diarrhea.

Calcium, which normally binds oxalate, binds fat instead, so free oxalate is absorbed by gut → ↑ frequency of calcium oxalate kidney stones.

Bilirubin

Heme is metabolized by heme oxygenase to biliverdin (green), which is subsequently reduced to bilirubin (yellow-brown). Unconjugated bilirubin is removed from blood by liver, conjugated with glucuronate, and excreted in bile.

Direct bilirubin: conjugated with glucuronic acid; water soluble (**d**issolves in water).

Indirect bilirubin: unconjugated; water **i**nsoluble.

